



RESEARCH ARTICLE

Power to the People! Assigning German Popular Music Genres Based on Listeners' Perspective Using a Genre Tree Approach

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Abstract

Despite the popularity of playlists on streaming platforms, music genres still remain a primary organizational tool among listeners, stakeholders, and researchers. Yet both 'correct' genre labels for individual popular music phenomena and structure of genre taxonomies are often contested. Nevertheless, assumed assignments often form the basis of music genre recognition models in MIR research as well as musical taste and recommender systems research, raising questions about reliability, validity, and comparability of results. By contrast, popular music studies have emphasized the listener-dependent nature of genre assignments. In other words: Genre assignment may be considered a matter of majority voting within a certain community. So why not make a virtue of necessity? We constructed a hierarchical tree from MusicBrainz folksonomy genre votes for 167,762 tracks listened to in Germany across the past 50 years. The resulting Community Music Genre Tree (COMGET) comprises 234 genres—usefully granular but probably too detailed for many research and applied tasks. We address this by an additional tree-folding algorithm that aggregates genres to the desired level of detail for any research problem. For automatic genre assignment of new tracks, we developed classifiers at multiple granularity levels using music features and meta-data drawn from online databases. Results indicate that our approach is able to assign genres that correspond to listeners' perspective, in a transparent, reproducible, and intersubjectively comprehensible way.

Keywords: music genre recognition, hierarchical taxonomy, folksonomy, network analysis, machine learning classification, German popular music

1. Introduction

Although playlist-driven listening has grown with the global diffusion of streaming platforms, genre labels still remain the primary way people and institutions organize popular music: Listeners search and browse by genre; playlist editors and DJs assemble genre playlists; A&R, marketing and label teams position and promote artists using genre tags; and music journalists and researchers rely on genre labels to describe and compare popular music (Bonini and Magaudda, 2024). Some scholars argue that classificatory practices have weakened since the late twentieth century (DiMaggio, 1987) and that certain genre boundaries are unstable (Silver et al., 2016), but empirical evidence shows that listening and social signaling remain organized around genres (Airoldi et al., 2016; Lonsdale,

2021). On most streaming platforms, genre assignments are embedded in metadata and pipeline logics—and hence used as search filters, playlist seeds, and features in recommender systems—which means platform algorithms amplify genres' practical significance for discovery and market positioning (Valverde, 2025). For researchers, conversely, genre labels serve both theoretical and methodological roles: they form variables in preference and reception studies (Lepa et al., 2020), inputs to recommender-system research (Porcaro et al., 2024), and useful aggregator variables for measuring perceived diversity and mitigating hubness in high-dimensional audio and metadata feature spaces (Flexer et al., 2018; Flexer and Grill, 2016).

However, 'correct' genre labels for individual popular music phenomena (i.e., tracks, artists, releases, scenes, market, playlists, labels, performances etc.) are frequently contested (van Venrooij and Schmutz,

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2018). Different actors may attach different genre labels to music releases for different reasons: artists employ them to signify their musical identity; publishers and distributors tag releases to support market placement; critics and musicologists apply normative or historical taxonomies; and listeners use genre terms to organize their listening collections and to negotiate their distinctive musical identity in everyday communicative interactions (Frith, 1996; Negus, 1999). Not only do chosen genre labels diverge, but the taxonomical systems employed also differ strongly in degree of differentiation. Commercial platform taxonomies illustrate this: For example, the commercial music metadata platform *Gracenote* uses a compact set of 25 genre tags (Jiang et al., 2024), whereas *Spotify* exposes many thousands of genre tags (McDonald, nd). Music databases and platforms also differ in terms of the objects to which they assign genres (track, release, or artist) and in assignment strategies (single primary genre versus multi-label tag sets).

These differences pose methodological challenges for empirical researchers across MIR and computational musicology. Problems include inconsistent genre labels that may degrade generalizability of supervised machine-learning models (Bogdanov et al., 2019), survey items on musical preferences that might be interpreted very heterogeneously by different listeners (Lange et al., 2025), and, on a larger scale, the issue of cross-study noncomparability that undermines replication and meta-analysis (Sturm and Flexer, 2023). Faced with these constraints, researchers often pragmatically adopt available commercial or open taxonomies, without systematically engaging with the actual source of labels or underlying theoretical assumptions (Green et al., 2024). To address these challenges, the present paper presents a reproducible procedure to derive a scalable and transparently constructed genre taxonomy for any given corpus of popular music based on the community votes of any genre folksonomy. Paired with algorithmic music-genre recognition, this approach can (1) surface empirical regularities and disagreements in genre label use that may inform popular music studies (Sturm and Flexer, 2023), and (2) provide a shared, probabilistic ‘currency’ for cross-disciplinary work in computational musicology and MIR—so that classifiers, preference studies, and sociological analyses may operate from a common, well-documented mapping between recordings and meta-genres.

1.1 Musical Genres as hierarchical discourse formation

In popular music, genres are organized hierarchically: Pachet and Cazaly (2000) identified genealogical links (e.g., *rock* as parent of *alternative rock*), geographical links (*hip hop* as supergenre of *German hip hop*), and historical-period links (*pop* as supergenre of *80s pop*) in existing taxonomies. As Brackett (2016, p.

8) argues, hierarchy depth is not limited to two levels but can extend virtually indefinitely—yet consensus on which labels to select and how to relate them remains contested. Expert-constructed taxonomies show limited agreement (Pachet and Cazaly, 2000), and expert classifications often diverge from listener community norms (Sordo, 2008). Adopting the power-critical perspective of cultural studies (du Gay et al., 2013), we interpret the discourse on genres in the popular music field through Foucault’s (2013) concept of *discourse formation*. From this viewpoint, artists, industry, and listeners, exert unequal influence over which genre labels circulate and stabilize on the long run. Given that in popular music emerging genres are often-times defined by listeners (Lena and Peterson, 2008), genre assignment can be understood as implicit majority voting within a community of listeners.

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We therefore propose a new methodological approach that derives a genre hierarchy and genre assignments for music tracks from the votings of global music listener communities, also termed *folksonomies*. Platforms such as *RateYourMusic* (nd), *Discogs* (2024), or *MusicBrainz* (nd) provide crowdsourced ‘ground truth’ on the genre of musical objects which is reflected in aggregated tagging behavior. *MusicBrainz* seems particularly suitable for our endeavour: it maintains a detailed whitelist of over 2,000 genre tags without imposing an a-priori hierarchy, and its democratic voting system allows users to assign and vote on tags at multiple levels (track, release, artist), enabling nuanced analysis of consensus and disagreement. Although we use *MusicBrainz* as our primary data source, external validation against other folksonomies (i.e., *Discogs*) appears necessary to judge generalizability (Sturm and Flexer, 2023). In a similar way, Schreiber (2015, 2016) demonstrated that hierarchical genre relationships can be inferred from tag co-occurrence patterns on digital platforms. Crucially, however, genre assignment conventions may vary across listener communities in different countries and time periods (Pichl et al., 2017; Nie, 2022). As a case study, we thus anchor our exemplary analysis in popular music tracks listened to in Germany over the past 50 years to ensure the resulting taxonomy reflects a coherent discursive context.

For the use of hierarchical genre taxonomies in empirical research, three pragmatic issues arise. First, balanced genre distributions improve statistical power and classifier performance (Japkowicz and Stephen, 2002). However, subgenres are typically assigned to fewer tracks than their super categories (Bogdanov et al., 2019), creating an inherent imbalance that must be considered when constructing or aggregating within the taxonomy. Second, the optimal number of genres depends on the research question: surveys of musical taste may require moderate to high resolution (Lange et al., 2025, e.g., 24 genres), robust machine learning

models often perform better with fewer, but broader categories (Genussov and Cohen, 2010), and fine-grained subcultural analyses may demand to differentiate many subgenres (Caparrini et al., 2020). At the same time, preserving consistency with the overall tree structure is desired. This implies to construct a scalable genre taxonomy that supports multiple resolutions derived from a single underlying hierarchy. Third, applying such taxonomies to large corpora in computational contexts requires an automated classifier that can assign genres in an efficient, consistent and transparent way, serving as a neutral arbiter when human dissent over labels is expected.

1.2 Towards an automated classifier for popular music studies

Music genre recognition is a core task in Music Information Retrieval. Green et al. (2024) provide a systematic state-of-the-art review and critique how genre theory from the social sciences and humanities is often overlooked when selecting features (for a systematic review on genre theories see Merlini, 2020). Commonly, researchers in MIR and music psychology assume that genre can be inferred from audio features alone (Tzanetakis and Cook, 2002; Rentfrow et al., 2011). This conflicts with a sociological approach to genres which we also adopt here and describes them as “systems of orientations, expectations and conventions that circulate between industry, text and subject” (Neale, 1980, p. 19), emphasizing extra-musical context alongside sound. Franco Fabbri’s (1982) genre theory offers such a synthesis: musical style (instrumentation, rhythm, harmony, timbre) and release milieu (subcultural context, artist positioning, market placement) together inform how listeners assign genres. Modern MIR tools can extract style features from audio signals (Bogdanov et al., 2013), and open music databases provide access to distributional metadata (label, chart performance, platform placement).

Lyrics occupy an intermediate position, conveying both emotional expression and stylistic content (Brand et al., 2019) as well as contextual information through language codes and topics that signal subcultural affiliation (Carbone et al., 2024). Further, a classifier might need to pay special attention to the *mainstream pop* or *pop* metagenre that Fabbri (1982) describes as encompassing eclectic, commercially optimized works that resist association with a specific milieu. In addition, artists from former niche scenes are often retrospectively relabeled as *pop* or *mainstream* after achieving commercial success (Holt, 2007, p. 20). Therefore, popularity itself seems to be a relevant feature dimension for classification. Lepa & Radke (XXXX) have thus recently proposed a feature space spanning four dimensions—musical style, release context, lyrical content, and popularity—to operationalize the multi-dimensional constitution of genre following

Fabri’s XXX approach.

Drawing on the theoretical and methodological foundations outlined above, the present work pursued three research questions:

RQ1 How can we construct an empirically grounded and sufficiently balanced hierarchical genre tree for popular music released in Germany?

RQ2 How can we aggregate the inferred tree structure to a scalable number of distinct supergenres suitable for machine learning and statistical applications?

RQ3 Will an algorithmic classifier be able to predict community genre assignments at different levels of detail with sufficiently robust performance employing stylistic, distributional, lyrical, and popularity features?

In the following sections, we will hence introduce a method to create a balanced hierarchical genre taxonomy based on genre co-occurrences in folksonomy tags, describe a folding algorithm to flexibly reduce the number of genres for various research tasks, present automatic classifiers for different numbers of supergenres, and finally discuss our findings and summarize the results.

2. RQ1: Balanced Hierarchical Genre Tree

2.1 Data – Popular Tracks in Germany (POPTRAG)

As initial demonstration of our new methodological approach, we perform the whole analysis depicted in this paper with a corpus of 254,334 unique popular tracks released and frequently listened to in Germany over the past 50 years (POPTRAG). Four sources comprise this corpus: 215,702 tracks from weekly *MediaControl* German single and album charts (18 Nov 1977–21 Nov 2024; reduced to 75% where linkage success to *Spotify* identifiers was possible); 26,949 *Spotify* recommendations from 126 German-market genre seeds (retrieved 29 Nov 2024; median 325 per seed, IQR 25, range 49–343); 8,801 tracks from 100 top-featured *Spotify* playlists for the German market (retrieved on XXXX, median 90 tracks per playlist, IQR 42, range 28–750); and, finally, 11,994 entries from *Germany’s Daily Top 200 Spotify* charts (01 Jan 2017–30 Nov 2024). From all sources, we excluded tracks shorter than 30 seconds (pertains to 853 tracks, likely non-musical snippets) and retained only unique title-artist combinations, preferring earlier release dates for duplicates.

In a second step, we then tried to acquire *MusicBrainz* track genre tags for all music tracks in the POPTRAG corpus. Hereto, only genre labels contained the *MusicBrainz* whitelist (snapshot 16 Jun 2025) were taken into account, reduced by tags referring to genres beyond popular music (*non-music*, *interview*, *comedy*, *nature sounds*, *children’s music*, *spoken word*, *standup comedy*). In case track-level tags were missing, we employed release tags instead. If these were missing, too,

we used the artist tags. In addition, we required any genre tag to appear with at least 50 distinct artists in the overall corpus to ensure sufficient discursive support for any genre, if not, they were discarded. Finally, we retained only tracks in the corpus that were exhibiting at least two community genre assignments fulfilling all of these criteria, leading to a removal of 86,572 tracks (34%) and yielding a *MusicBrainz*-tagged subset of $N = 167,762$ tracks annotated with 234 distinct music genre tags spanning release years 1924–2024 (median 2012, IQR 16 years). Community genre assignments comprised between 1 and 22 different tags per track (median 4, IQR 3) and cast between 2 and 135 votes per track (median 6, IQR 7).

2.2 Procedure: From Genre-Tag Co-Occurrences to an Hierarchical Genre Tree

Based on the denoised German POPTRAG corpus, we algorithmically built a hierarchical genre tree from *MusicBrainz* tag co-occurrences. Thereto, we first constructed a weighted co-occurrence network where nodes represented *genre* labels and directed edges $e_{A \rightarrow B}$ quantified how strongly *genre A* predicted *genre B* on the same track. For each ordered pair (A, B) , we computed a vote-weighted conditional frequency:

$$e_{A \rightarrow B} = P(B | A) \times \frac{1}{|T_{A,B}|} \sum_{t \in T_{A,B}} \frac{v_{t,B}}{v_{t,\text{total}}} \quad (1)$$

where $T_{A,B}$ denotes tracks tagged A and B , $v_{t,B}$ the votes for B on track t , and $v_{t,\text{total}}$ the total votes on t . This formula produced asymmetric edges ($e_{A \rightarrow B} \neq e_{B \rightarrow A}$) that combined directional co-occurrence with community vote strength. To illustrate: suppose 1,000 tracks carried *metal* and 600 also carried *rock*, giving $P(\text{rock}|\text{metal}) = 0.6$. If the average vote share for *rock* on co-tagged tracks equalled 0.5, then $e_{\text{metal} \rightarrow \text{rock}} = 0.6 \times 0.5 = 0.3$. Conversely, 6,000 tracks carried *rock*, 600 of those carried *metal*, so $P(\text{metal}|\text{rock}) = 0.1$; with an average *metal* vote share of 0.7 on *rock* tracks, $e_{\text{rock} \rightarrow \text{metal}} = 0.07$. These weights express both prevalence and tag confidence.

In a second step, we resolved reciprocal track connections to obtain a unidirectional scaffold. For each pair, we thereto subtracted the weaker directional probability from the stronger and kept only the positive residual:

$$e'_{A \rightarrow B} = (e_{A \rightarrow B} - e_{B \rightarrow A})_+ \quad (2)$$

$$e'_{B \rightarrow A} = (e_{B \rightarrow A} - e_{A \rightarrow B})_+ \quad (3)$$

In the example this yields $e'_{\text{metal} \rightarrow \text{rock}} = 0.6 - 0.07 = 0.53$ and $e'_{\text{rock} \rightarrow \text{metal}} = 0$, making *metal* $\rightarrow$ *rock* a clear candidate supergenre link.

Third, we enforced a strict tree-like structure by assigning each node its single strongest outgoing edge as

its *supergenre*; all other edges were discarded. Components without *supergenre* were attached to an artificial root *POPULAR MUSIC* to guarantee connectivity. In the example, *metal* would attach to *rock* unless a stronger edge existed, and *rock* would attach to the root if it had no *supergenre*.

Finally, we balanced the tree in terms of optimizing leaf-node populations on each tree level. Thereto, we estimated each genre node's size as the sum of its assignment probabilities across tracks in the corpus, then computed the Gini coefficient within each hierarchical tree-level for all genre nodes contained, and defined overall tree imbalance as the weighted mean of all tree-level Ginis. We then searched for the least-imbalanced tree by iteratively removing the weakest hierarchical link of the whole tree, reattaching the resulting dangling tree to *POPULAR MUSIC*, and recomputing imbalance again; we then finally selected the iteration with minimal overall tree imbalance.

2.3 Results: Community Music Genre Tree (COMGET)

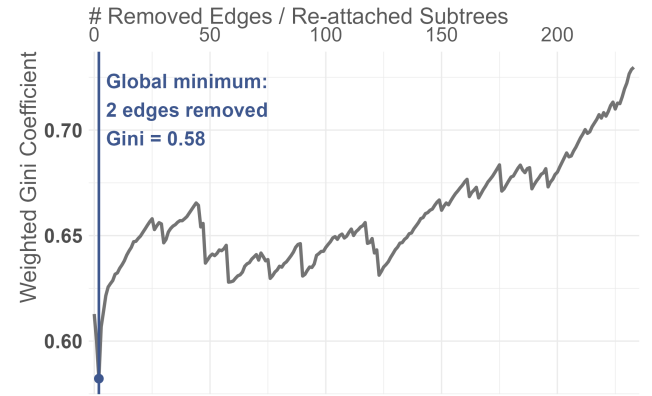


Figure 1: Weighted Gini coefficient (0–1) across iterations of cutting weak genre links and reconnecting dangling subtrees to *POPULAR MUSIC*.

Level	# nodes	# tracks	Gini
1	3	total: 47,846 (29%)	0.136
		12,472; 16,401; and 18,973	
2	119	total: 90,381 (54%)	0.643
		median: 263 IQR: 789 range: 10.4 to 5832	
3	110	total: 29,343 (17%)	0.531
		median: 156 IQR: 259 range: 14.7 to 1496	
4	2	total: 192 (0.1%) 55.6 and 136	0.420

Table 1: Structure of COMGET at different hierarchy levels; track counts are sums of genre tag assignment probabilities in POPTRAG.

The initial POPTRAG *genre* co-occurrence network contained 234 *genre* nodes forming a single connected component with 45.14% density, indicating relatively high inter-tag connectivity. By median, *genres* co-occurred with 97 different others (IQR 63.5; range 3 for *waltz* to 229 for *pop*), revealing large differences between niche and ubiquitous tags. After resolving reciprocal directions, each *genre* received on average 29.5 incoming subgenre links (IQR 32; min 0; max 228 for *pop rock*) and 55 outgoing supergenre links (IQR 32; min 0 for *rock*; max 105 for *dub*).

The balance optimization procedure (Figure 1) reattached *pop* and *hip hop* directly to *POPULAR MUSIC*—*pop* had formerly been subordinate to *rock*, and *hip hop* to *pop*, however these connections where the weakest in the whole tree and their cutting resulted in clear improval of overall balance. The three weakest remaining links are *electronic* to *pop* (0.22), *classical* to *pop* (0.27), and *jazz* to *pop* (0.29). The final *Community Music Genre Tree* (COMGET) is highly leaf-heavy: 89.36% of *genre*-nodes lack subgenres. *Rock* emerges as the largest *supergenre* with 63 direct *subgenres*, and only *rock*, *pop*, and *hip hop* remain connected directly to *POPULAR MUSIC*, forming three broad meta-categories. The tree exhibits four hierarchy levels (excluding the artificial root); see Table 1 for structural details. An interactive visualization of COMGET is available in the digital appendix.

3. RQ2: Scalable Granularity

3.1 Methods: Folding Algorithm

In order to derive a scalable number of *genre* categories suitable for diverse research questions and statistical applications from COMGET, we developed a greedy folding algorithm aiming at three requirements: optimal *genre* category balance, a user-selectable number of *genre* categories, and sufficient track counts per *genre* for robust machine learning and statistical analysis. The algorithm iteratively merges the least populous *genre*-nodes furthest from the root of the COMGET tree with their *supergenres* until all remaining *genres* meet a minimum track count threshold. Here, *genre node size* is again computed as the sum of vote-weighted assignment probabilities across all tracks. For a desired number of *genre categories*, users can now select the folding solution corresponding to the local Gini minimum, drawing on a tree-level-weighted Gini, thereby ensuring well-balanced categories for each original tree level.

After folding COMGET to the desired number of *genre* categories, each track in the corpus finally obtains a single *genre* label via majority voting: all valid original *MusicBrainz* tags are assigned to a track based on their *category assignment* in folded COMGET, then each track receives the *category* tag with the highest summed votes. Ties are resolved by selecting the *genre* with lower total assignment probability in the

full dataset, thereby favoring more detailed *genre* assignments. Please refer to the digital appendix for a more detailed description of the folding and *genre* assignment algorithm.

For the present exemplary analysis using the POPTRAG corpus containing German popular music, we conducted a grid search across minimum *genre sizes* from 1,000 to 15,000 in increments of 5, maximizing within-level balance as measured by a level-weighted Gini coefficient. We then chose 4 different local minima from results for comparison. For demonstrating the content validity of our exemplary 10-*genre* solution for the German popular music market derived from COMGET, we compared resulting *genre* category assignments to the *genre* labels assigned by the *Discogs* community using a contingency table.

3.2 Results: Germany's Popular Music Supergenres

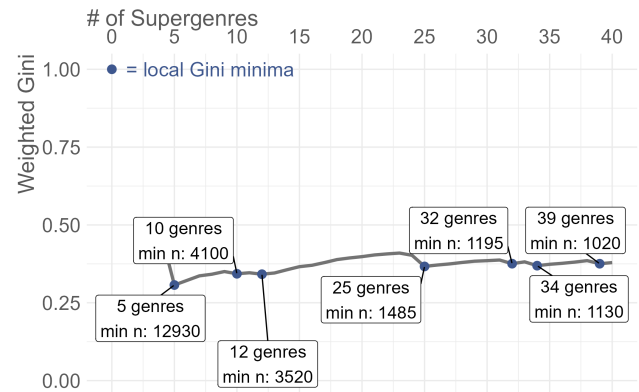


Figure 2: Weighted Gini coefficient (0–1) across COMGET folding solutions for minimum track counts from 1,000 to 15,000 in increments of 5, using the POPTRAG corpus.

For our POPTRAG corpus, the folding algorithm identified multiple optimal solutions for a German COMGET (Figure 2): a low-resolution solution with 5 *genres* (min. 12,930 tracks, Gini 0.307), medium-resolution solutions with 10 and 12 *genres* (min. 4,100 and 3,520 tracks; Gini 0.343 and 0.342), a high-resolution solution with 25 *genres* (min. 1,485 tracks, Gini 0.367), and finer solutions with 32, 34, and 38 *genres* (min. 1,195, 1,130, 1,020 tracks; Gini 0.376, 0.369, 0.375). We focused subsequent analyses on solutions with 5, 12, 25, and 32 *genres*, representing different granularities suitable for various research questions (Figure 3).

COMGET-5G comprises *rock*, *pop*, *hip hop*, *electronic*, and *metal* (in descending order of track count). In COMGET-12G, *jazz* and *schlager* emerge as further distinct *pop subgenres*, while *rock subgenres* including *heavy metal*, *alternative rock*, *pop rock*, *hard rock*, and *indie rock* appear at the same level. Notably, listeners do not classify *heavy metal* as a *subgenre* of *metal* but as a direct *subgenre* of *rock* on the same hierarchy level

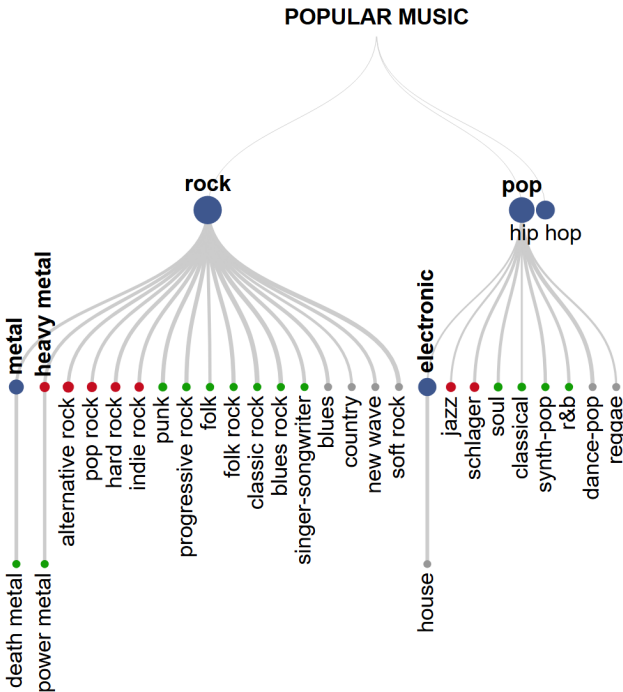


Figure 3: Folded COMGET hierarchies for POPTRAG corpus at four different resolutions (5, 12, 25, 32 genre categories). Colors indicate additional genres introduced at each finer resolution (blue = 5; red = 12; green = 25; grey = 32). Node size represents approximate track count in POPTRAG (sum of vote-weighted assignment probabilities).

as metal.

COMGET-25G introduces further *rock subgenres* (*punk, progressive rock, folk, folk rock, classic rock, blues rock, singer-songwriter*) and additional *pop subgenres* (*soul, classical, synth-pop, r&b*). A third hierarchy level appears with *death metal* as a *subgenre of metal* and *power metal* as a *subgenre of heavy metal*. COMGET-32 adds *house* as an *electronic subgenre* and additional *rock and pop branches* (*blues, country, new wave, soft rock, dance-pop, reggae*). *hip hop subgenres* appear only in more fine-grained solutions (*pop rap* as 33rd genre, *trap* as 38th genre).

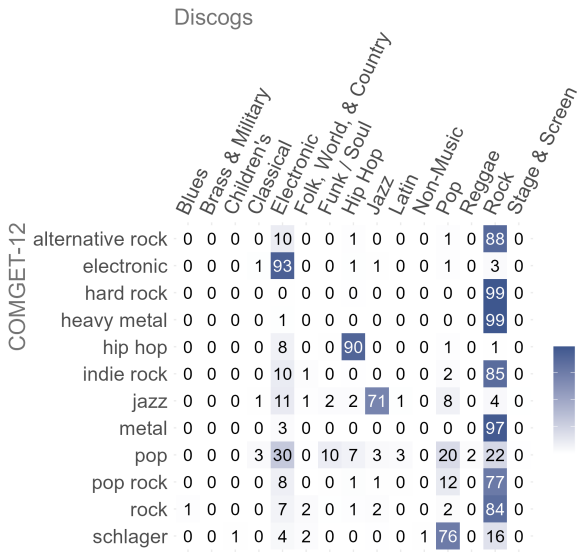


Figure 4: Contingency table comparing COMGET-12G and Discogs genre assignments; values show $P(\text{Discogs} \mid \text{COMGET-12})$.

Discogs' 15-genre taxonomy approximates COMGET-12 in scale. For approximately 5% of tracks with COMGET assignments, we found no *Discogs* counterpart, introducing minimal popularity bias. The contingency table (Figure 4) reveals strong alignment for *electronic, hard rock, and metal*. Even where *Discogs* includes a *Jazz* category, COMGET's *jazz* overlaps substantially with *Discogs'* *Electronic* and *Pop*. Similarly, *pop rock* is distributed across *Discogs'* *Rock, Pop, and Electronic*; *schlager* emerges in *Pop* (similar label to its COMGET *supergenre*) with minor overlap to *Rock*. Five *Discogs* categories—*Blues, Brass & Military, Children's, Non-music, and Stage & Screen*—show minimal correspondence to any COMGET-12 genre, mostly consistent with our noise-reduction filtering and focus on popular music. COMGET's *pop* category is distributed widely across *Discogs genres* (*Electronic, Pop, Rock*), reflecting pop's eclectic character. Seven of twelve COMGET-12 genres overlap most strongly with *Discogs'* *Rock*, suggesting *Discogs* lacks granular rock subgenre labels needed to represent German popular music adequately.

4. RQ3: ML-Classifier

4.1 Methods: Outcome, Features, Workflow

To determine whether an automated classifier can predict the *supergenres* identified by our COMGET folding algorithm, we derived *single genre tags* for each track at four levels of detail (COMGET-5G, COMGET-12G, COMGET-25G, COMGET-32G) and trained independent classifiers for each. Thereto, we augmented POPTRAG with 225 descriptor variables spanning *musical style*, *release context*, *lyrical content*, and *popularity* from music platforms *Spotify*, *MusicBrainz*, *AcousticBrainz*, *Genius*, and *Deezer* via ISRC- or fuzzy string matching employing either API-access or web scraping. *AcousticBrainz* provided community-calculated *audio features*; *MusicBrainz* supplied artist demographics (*persona*, *origin*, *birth year*, *gender*); *Spotify* and *Deezer* contributed *popularity* metrics and *Echo-Nest audio features*. From *Spotify's* artist-level genre tags, we derived 58 *distribution genre* flags (e.g., “Is a track’s artist tagged as ‘reggae’ on *Spotify*?”) using the same co-occurrence algorithm applied to COMGET, but operating on *Spotify* artist genres as input. *Distribution genres* capture how the music industry positions and markets artists within commercial taxonomies, which may diverge from listener perception due to strategic positioning, legacy labelings, or cross-genre appeal (Valverde, 2025). By including these industry-assigned categories alongside audio and lyrical features, we provide the classifier with a signal of release milieu that complements community-based genre assignments and tests whether industry framing influences how listeners ultimately categorize music.

In our workflow, we first discarded observations with more than 40% missing values on any of the variables, retaining 155,093 tracks (92%), for which we imputed missing values by medians. We performed an 80/20 train-holdout split at the artist level to mitigate artist leakage effects (Flexer and Schnitzer, 2010), and used 5-fold cross-validation (also group-splitted based on artists) for hyperparameter tuning. We lumped categorical variable levels with fewer than 100 observations in any fold into an “other” category, then discarded variables that contained only this category. All remaining categorical predictors were dummy-coded with the most common level as reference.

We first evaluated four different supervised ML algorithms on a 20% subsample at low and medium resolution to balance prediction performance and interpretability: elastic net (GLMNET), regularized discriminant analysis (RDA), random forest (RF), and light gradient boosted machine (LightGBM). For each, we tuned key hyperparameters: GLMNET (*penalty*, *mixture*), RDA (*covariance fraction* and *identity fraction*), RF (*split variables*, *leaf node size*, *tree depth*; *number of trees*: 500 for tuning, 1,000 for final models), and LightGBM (*boosting rounds*, *learning rate*, *leaf count*, *tree depth*, *feature fraction*). We tuned hyperparam-

eters across k-folds using the macro-averaged F1 score (mean per-class F1, set to 0 for unpredicted classes) to treat all *genres* equally during evaluation. Preprocessing was applied in sequence within each single fold: median imputation, adaptive class balancing (excluded from holdout set), zero-variance removal, and normalization. Adaptive balancing used a single tunable *target-ratio* parameter governing rebalancing degree. The algorithm computed a reference count from class frequencies (median for ≤ 10 classes; 60th percentile for 11–20 classes; 65th percentile for > 20 classes), then downsampled classes exceeding reference $\times$ *target ratio* using NearMiss (Zhang and Mani, 2003) and upsampled classes below reference $\div$ *target ratio* using Borderline SMOTE (Han et al., 2005) or regular SMOTE when sufficient danger observations were unavailable. This approach preserved more data than fixed-ratio sampling while remaining tunable: *target ratio* = 1 enforced perfect balance; higher values retained original imbalance. The reference-count heuristic scaled to problem complexity, preventing excessive downsampling at higher resolutions.

For identifying best hyper-parameters, we employed Bayesian optimization (Snoek et al., 2012) with a Latin hypercube initial grid (20 points for full analysis, 10 for prototyping) (Husslage et al., 2011), a maximum of 50 Bayes steps (25 for prototyping), early stopping if no improvement occurred after 10 iterations, and exploration of uncertain hyperparameter regions after 5 iterations without improvement. Hyperparameters were then selected using the one-standard-error rule (Hastie et al., 2009). After identifying them, we trained final models on the full training set, and evaluated results on the holdout set. *Community certainty* served as a reference point for evaluating final classifier performance, since ML models cannot exceed the consistency of the underlying human community assignments. For this comparison *Certainty* was computed as the mean assignment probability across votes for each track assigned to the *single-label genre*. All random operations used the seed 42 for reproducibility.

4.2 Results: Classification Performance for different degrees of granularity

Model comparison (Figure 5) shows that all four learners perform similarly on the 20% subsample in terms of macro F1. LightGBM emerged as the strongest performer at both resolutions (COMGET-5G and COMGET-12G) and was selected as the final learner for all subsequent analyses. GLMNET ranked second for COMGET-5G but performed worst for COMGET-12G, indicating that its linear decision boundary assumption does not scale to higher-resolution classification tasks.

The final GBM models on the full dataset (Table 2) achieved $F1_{\text{macro}} = 0.803$ (SE: 0.008) on CV folds and 0.788 on the hold-out set for COMGET-5G, indicating slight overfitting. Per-class F1 correlates strongly with

# genres	boosting rounds	tree depth	feature fraction	leaf count	learning rate	target ratio
5	854	4	83	98	0.049	1.177
12	846	10	35	4	0.037	1.106
25	898	5	48	67	0.006	2.394
32	854	4	83	98	0.049	2.462

Table 2: Hyperparameter settings for final LightGBM models at different granularity levels.

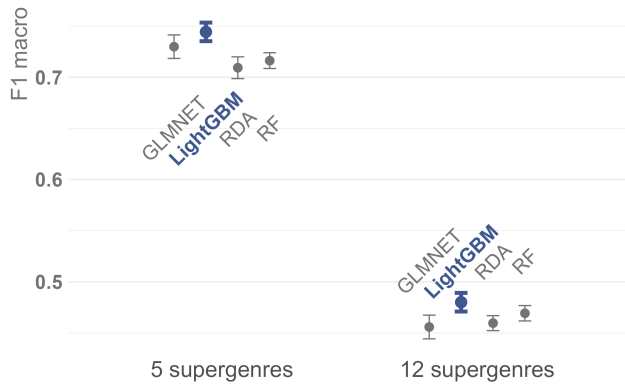


Figure 5: Performance comparison of different learners (GLMNET, RDA, RF, GBM) on 20% of the data set for COMGET-5G (left) and COMGET-12G (right); plots show mean and standard error of $F1_{macro}$ across CV folds.

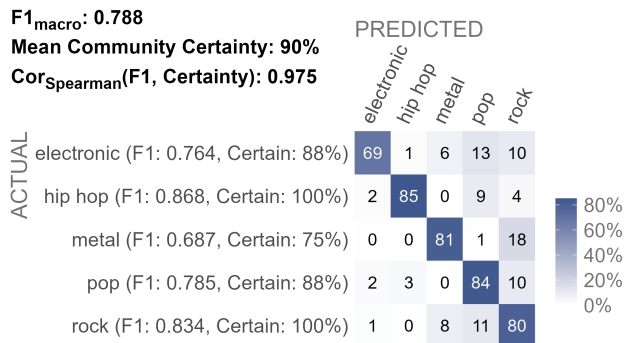


Figure 6: Confusion matrix for the final GBM model on the hold-out set for COMGET-5G. Values show $P(\text{Predicted}|\text{Actual})$; *certainty* indicates the mean assignment probability for each *genre* across community votes.

community assignment certainty ($\text{Cor}_{\text{Spearman}} = 0.975$). Given mean community certainty of 90%, the classifier performs satisfactorily. Cohen's $\kappa = 0.723$ on the hold-out set ($\kappa_{CV} = 0.734 \pm 0.009$) meets established interrater agreement standards in the social sciences (Hildebrandt et al., 2015, p. 327). The confusion matrix (Figure 6) reveals the model struggles most with *electronic*, often confusing it with *pop* and *rock*. Electronic music's stylistic elements frequently appear across pop and rock subgenres, explaining the observed misclassifications. Similarly, *metal* misclassifications with *rock* reflect their close relationship.

COMGET-12G performance degraded to $F1_{macro; CV} = 0.572 \pm 0.013$ and $F1_{macro; hold-out} = 0.599$ ($\kappa_{CV} = 0.618 \pm 0.010$; $\kappa_{hold-out} = 0.623$), with lower community certainty (67%), indicating greater difficulty for both community and classifier. Per-class F1 and *certainty* remain highly correlated ($\text{Cor}_{\text{Spearman}} = 0.831$). The confusion matrix (Figure 7a) shows primary confusion among *rock subgenres* (*alternative rock*, *hard rock*, *indie rock*, *pop rock*). *Heavy metal* is frequently confused with *metal* but rarely vice versa, and sometimes with *hard rock*, despite being recognized as distinct from *rock*. *Indie rock* and *jazz* are often misclassified as *electronic*; *jazz* also confused with *pop*. *pop rock* performs worst ($F1 = 0.160$), matching its low community certainty (50%). It is primarily confused with *rock* ($F1 = 0.670$, certainty = 75%) and *pop* ($F1 = 0.739$, certainty = 77%), suggesting *pop rock* shares ambiguous characteristics with both. Conversely, *schlager* and *hip hop* tracks are classified quite well ($F1 = 0.837$ and 0.858 , respectively) despite differing community certainties (67% and 100%), indicating these *genres* possess very distinct characteristics in the feature space.

COMGET-25G showed further loss in classification performance ($F1_{macro; CV} = 0.419 \pm 0.005$; $F1_{macro; hold-out} = 0.425$; $\kappa_{CV} = 0.486 \pm 0.010$; $\kappa_{hold-out} = 0.469$) alongside decreased community certainty (56%). The correlation between *certainty* and F1 weakened ($\text{Cor}_{\text{Spearman}} = 0.662$), yet remained high, suggesting other factors influence performance at higher granularity. The confusion matrix (Figure 7b) again reveals persistent *rock subgenre* confusion (*alternative rock*, *blues rock*, *classic rock*, *folk rock*, *pop rock*) and *metal*-related confusion (*death metal*, *folk rock*, *hard rock*, *heavy metal*, *power metal*). A secondary confusion center emerges around *pop*, which frequently misclassifies with *alternative rock*, *classical*, *electronic*, *folk rock*,

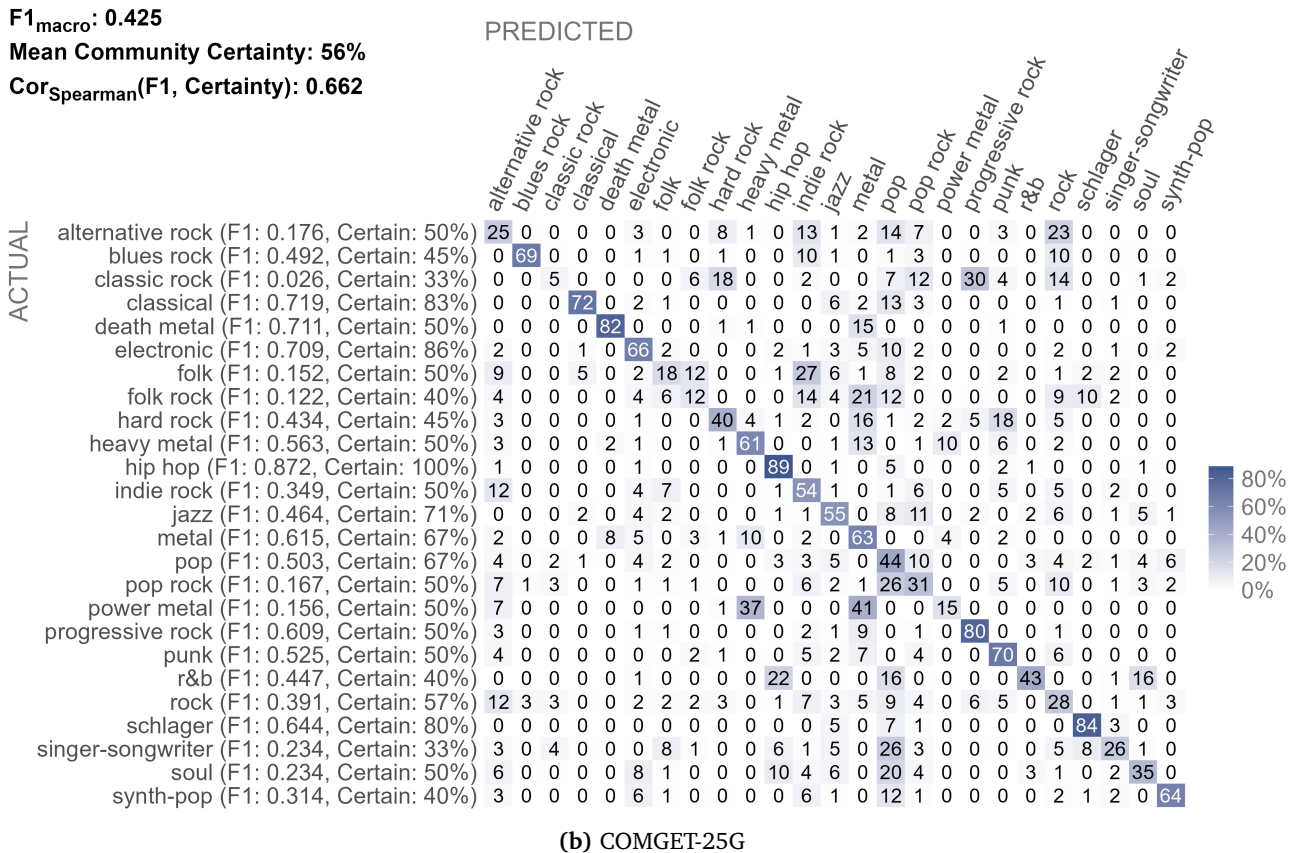
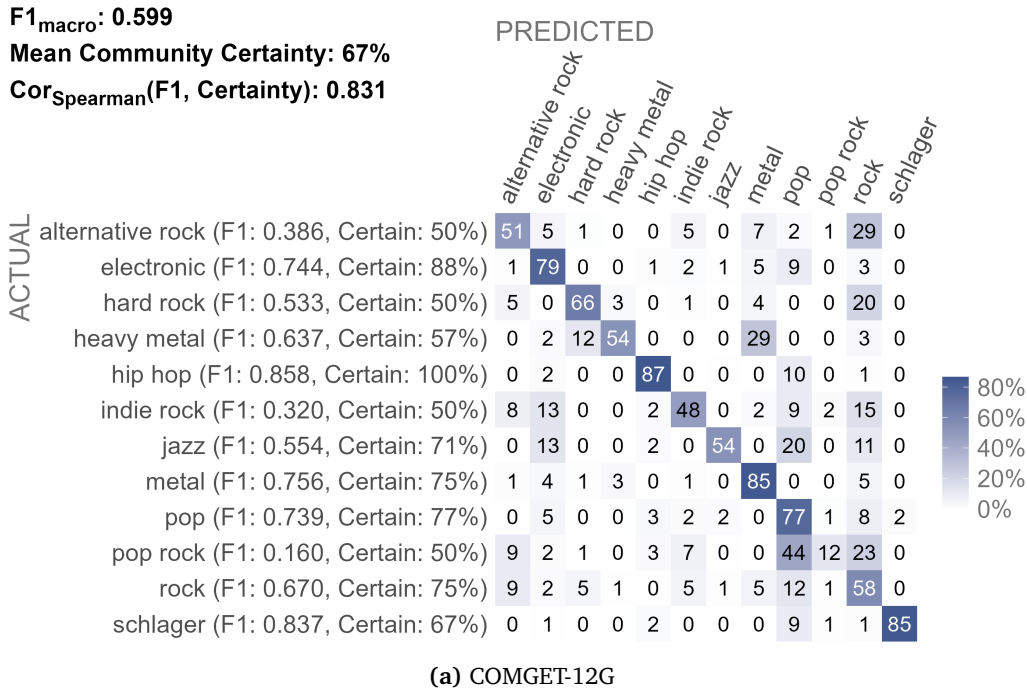


Figure 7: Confusion matrices for the final GBM models on the hold-out set. Values show $P(\text{Predicted}|\text{Actual})$; *certainty* indicates the mean assignment probability for each *genre* across community votes.

pop rock, r&b, singer-songwriter, soul, and synth-pop. The model performs well ($F1 > 0.7$) on classical, death metal, electronic, and hip hop—genres with distinct feature signatures and moderate to high community certainty (ranging 50%-100%). Poor performers (alternative rock, classic rock, folk, folk rock, pop rock, power metal with $F1$ ranging 0.026-0.176, certainty ranging 33%-50%) represent fuzzy genres difficult to classify for both community and model.

COMGET-32G yielded a further degradation in benchmarks ($F1_{\text{macro}}; CV = 0.400 \pm 0.003$; $F1_{\text{macro}}; \text{hold-out} = 0.403$; $\kappa_{CV} = 0.477 \pm 0.005$; $\kappa_{\text{hold-out}} = 0.486$) with decreased certainty (54%). Importantly, the correlation between certainty and $F1$ recovered notably ($\text{Cor}_{\text{Spearman}} = 0.795$), indicating community uncertainty becomes the dominant performance factor at extreme granularity. Given similarity to COMGET-25G, detailed confusion matrix results appear in the digital appendix.

Figure 8 shows the top 20 most important features for automated classification of COMGET-5G, COMGET-12G, and COMGET-25G with the final LightGBM learner. While we find similar features in the top 20 across all three levels of granularity (e.g., *acousticness* (ECHOEST), *artist distributed as “metal”*, *age of artist*), the importance of single features differs. For COMGET-5G, features that describe general style characteristics (e.g., *acousticness*, *speechiness*, *explicitness of lyrics*) and artist persona (e.g., *single artist vs. band*, *age of artist*) are most important. For COMGET-12G, popularity features and the *release year* gain importance. For COMGET-25G the most important features are mostly artist-related, indicating that fine-grained genre distinctions rely primarily on release-milieu signals rather than audio properties.

5. Discussion

With the present article, we introduce a novel and methodologically transparent procedure for constructing hierarchical genre taxonomies and according genre category assignments from listener-community discourse contained in folksonomies on open music database. Specifically, we make three key contributions: (1) an algorithm to construct a balanced hierarchical genre tree (COMGET) from folksonomy genre tag co-occurrences, (2) a folding algorithm that generates optimally balanced hierarchical genre category systems and final track-assignments at user-specified granularities from such trees, and (3) we demonstrate that resulting community-based genre assignments are a) comparable to existing open genre taxonomies (i.e. Discogs) and b) can be learned well with supervised machine learning by drawing on openly available music descriptors, hence can be used as a transparent, reproducible, and empirically grounded alternative to commercial or expert-curated genre taxonomies in future applied or basic research. We demonstrated our

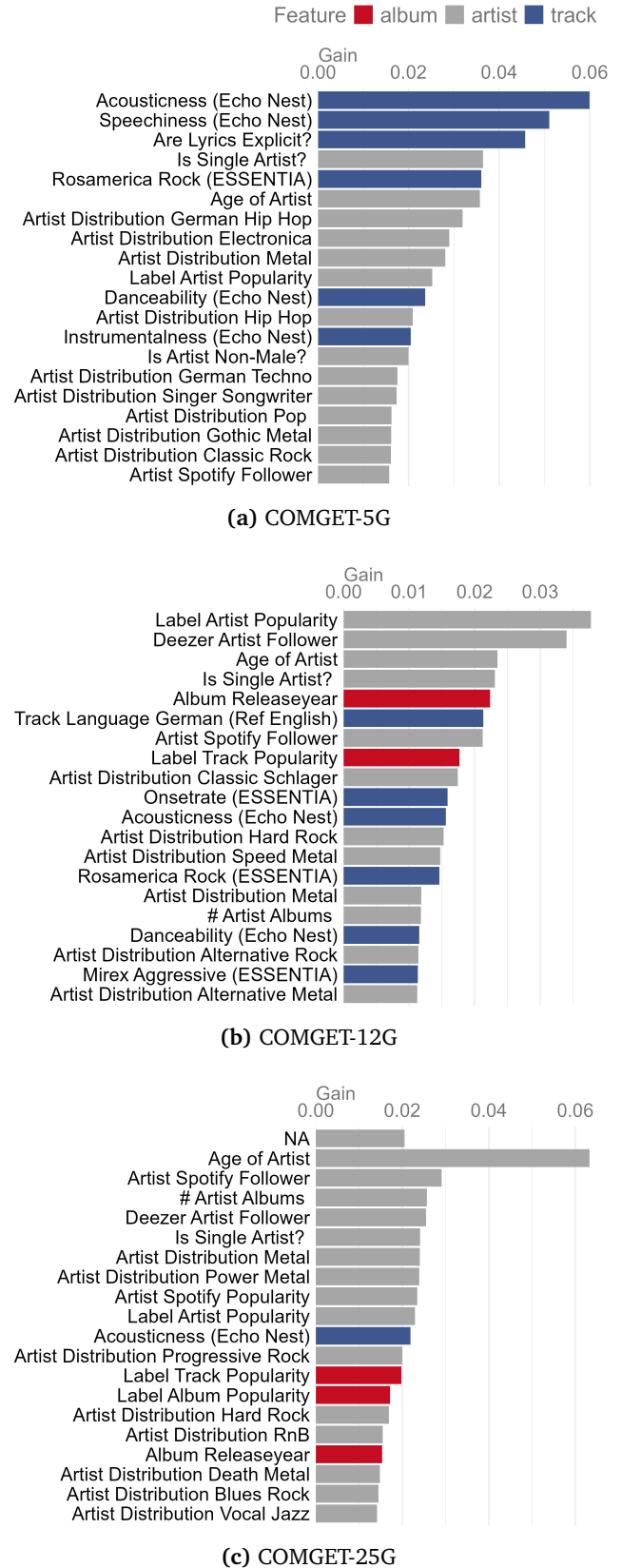


Figure 8: Top 20 most important features for automated classification at three granularity levels. Values show relative feature importance from the final LightGBM models.

approach with a large German popular music corpus (POPTRAG) and drawing on *MusicBrainz* folksonomy votes and the POPTRAG corpus. Nevertheless, it can in principle be adopted to any other music corpus and other genre tagging systems that allow co-occurrences of tags (e.g. Spotify, YouTube, SoundCloud).

5.1 Balanced and Scalable Representation of Discourse

While our approach shares some similarity with Schreiber’s algorithm (Schreiber, 2015) we grounded our labels exclusively in a single folksonomy rather than mixing commercial taxonomies with community votes. This choice reflects our power-critical perspective from cultural studies that we have discussed in the introduction and (du Gay et al., 2013) that is often adapted in popular music studies. Our method thus addresses both MIR methodology issues as well as ongoing humanities and social science debates about proper cultural classifications (DiMaggio, 1987). The principle of democracy operates at multiple levels in our approach: genre labels and co-occurrences rely on listener votes, and our algorithmic pipeline embeds democratic decision-making throughout; we selected the strongest hierarchical link for each supergenre assignment and employed majority voting to convert multi-label tags into single labels for classifier training.

Furthermore, our approach adheres directly to practical requirements for the use of genre assignments in social research and MIR. It generates balanced hierarchies without having to set arbitrary thresholds or hyperparameters, making it portable to any corpus with multi-tag genres. In fact, we demonstrated this by inferring *distribution genres* from *Spotify* artist tags in RQ3 of this paper. Crucially, our folding algorithm is able to derive optimally balanced solutions at different scales using the same statistical criterion (weighted Gini coefficient) that balanced the full taxonomy, enabling researchers to compare research results on various levels of genre granularity while maintaining a consistent conceptual framework. Beyond academia this approach might be also of interest for industry stakeholders such as radio stations and streaming services needing to organize large music collections in multiple granularities.

We would like to emphasize that COMGET taxonomies do not intent to represent historical music genealogies as Pachet and Cazaly (2000) conceptualized them. Rather, they depict how listeners hierarchically organize popular music discourse, i.e., which genres they relate to one another. Genealogical links such as *rock* as supergenre of *alternative rock* appear, but so do counterintuitive ones: *classical* and *jazz* emerge as *pop* subgenres. From our perspective, this outcome rather reflects corpus composition in interaction with folksonomy logic. POPTRAG mostly comprises accessible and exceptionally popular tracks, e.g. from crossover artists, popular opera arias, soundtracks, popular jazz

singers, or smooth jazz. Combined with the very frequent use of broad labels (Bogdanov et al., 2019) this leads to the observed hierarchy. Similarly, both *heavy metal* and *metal* appear as direct *rock* subgenres, explained by heavy metal’s roots in late-1970s psychedelic rock and hard rock alongside metal’s later distinctiveness (Gracyk, 2016).

Rock, pop, and hip hop emerged as three primary genre clusters in German popular music. These findings coincide with Silver et al.’s (2016) observation of three main genre clusters within international popular music (based on 3 million artists on *MySpace*, 2007). In line with their observations, Hip hop stands apart: it shows minimal subgenre differentiation, strong coherence around its supergenre, and fewer subgenres than rock or pop clusters. However, while Silver et al. (2016) distinguished between “Rock’n Roll” and “Niche” genres, we find that rock genres form a distinct cluster in popular German music, while pop genres are diverse in terms of musical styles, awith niche genres scattered across all three clusters.

5.2 Classifier Performance in MGR

As (Holt, 2007, p. 5) noted, semantic codes of musical genres are fragile. This makes MGR such a challenging endeavor. Even experts in the field do not agree completely on genre assignments (P’almason et al., 2018); so apparent misclassifications might also reflect genuine genre overlap rather than model failure. In other words: Models cannot exceed human consistency (Sturm and Flexer, 2023). Given community uncertainty in genre assignments, our classifiers achieved reasonable overall performance. We observe the expected degradation with increasing label granularity (Genussov and Cohen, 2010), mirroring declining community certainty. Also, subgenres are harder to be detected than supergenres in general, probably due to shifting boundaries (Silver et al., 2016). Artist-level features dominated across granularities, indicating that release milieu (i.e., how platforms and industry position artists) guides listener genre perception. This aligns with sociological genre theory emphasizing industry, text, and subject as coconstitutive (Neale, 1980). Stylistic features, such as audio and lyrics features, contributed significantly but secondarily. Lyric features especially showed weak predictive power, despite utility in other work (Mayer et al., 2008). Incorporating LLM-based lyric embeddings might capture cultural context more richly and better discriminate release milieu (Prevedello et al., 2024).

5.3 Limitations and Future Research

Several constraints limit generalization. *MusicBrainz* launched in 2000 without individual vote timestamps, so COMGET aggregates roughly 25 years of listener votes rather than reflecting a single moment; genre boundaries and perception likely shifted within this window. Platform user demographics and expertise re-

main opaque (approximately 1.2 million anonymous editor accounts; MusicBrainz nd), preventing assessment of demographic or expertise bias. Popularity bias affected coverage: 36% of initial tracks received fewer than two MusicBrainz votes, potentially underrepresenting niche or rare recordings. However, since we target Germany's popular music genres, this constitutes only a minor limitation. Methodologically, collinearity among metric predictors complicates feature-importance interpretation and may have caused instability in which features appeared predictive across models.

Future work should pursue three directions. First, convene popular music experts to review the taxonomy, label exemplar tracks, and identify misassignments to strengthen COMGET's validity. Second, replicate the pipeline on independent folksonomies or regional corpora of tracks (e.g., US and UK charts) to reveal how national listening cultures reshape genre structure. Third, enhance classifier prediction through advanced lyric features or learners.

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